Fault-Tolerant Quantum Error Correction at Room Temperature: Noise Model Selection and GNN Decoder Advantage for Continuous-Variable Photonic Architectures

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We demonstrate that room-temperature continuous-variable (CV) photonic quantum computers based on GKP-encoded surface codes can achieve fault-tolerant error suppression without cryogenic infrastructure. Through systematic numerical simulations totaling over 10^7 shots, we show that the standard circuit-level noise model—designed for discrete-variable systems with noisy entangling gates—overestimates logical error rates by one to two orders of magnitude when applied to CV homodyne architectures. We identify the phenomenological model with equal data and measurement error rates ($p_{\text{meas}} = p_{\text{data}}$) as a physically motivated noise model for CV systems, where homodyne detection always succeeds and the sole noise source is GKP displacement noise propagated through passive beam-splitter networks. Furthermore, we establish that soft-information minimum-weight perfect matching (MWPM) decoding is not merely an optimization but a necessity: it provides over an order of magnitude improvement in multi-round simulations and determines whether the system is above or below threshold. At the operating parameters of a room-temperature PPLN OPA photonic system ($\sigma_{\text{eff}} = 8.5$ dB), our CV-motivated model yields $p_L(d{=}5) = 4.33 \times 10^{-4}$ (13 errors in 30,000 shots, 95% CI: $[2.5, 7.3] \times 10^{-4}$), well below the 10^{-3} product threshold. Finally, we show that a lightweight graph neural network (GNN) decoder with 4,033 parameters outperforms soft-info MWPM by up to 14× at $d = 7$ under correlated noise ($\rho \geq 0.08$), with the advantage *increasing* monotonically with code distance (1.9× at d=3, 7.8× at d=5, 14× at d=7)—a regime where MWPM performance degrades. These results establish that cryogenic infrastructure is not a fundamental requirement for fault-tolerant quantum computing, and that ML-based decoders can extend the fault-tolerance operating envelope of CV photonic systems.

I. INTRODUCTION

A. The cryogenic bottleneck

Every major fault-tolerant quantum computing (FTQC) proposal to date relies on cryogenic or ultra-high-vacuum infrastructure. Superconducting platforms operate at ~ 4 mK with dilution refrigerators consuming tens of kilowatts [1, 2]. Trapped-ion systems require $\sim 10^{-11}$ torr vacuum chambers [3]. Even photonic proposals based on discrete-variable (DV) encoding demand single-photon detectors operating at ~ 0.8 K [4, 5]. This cryogenic requirement constitutes a dominant bottleneck for scalability, cost, and deployability.

A natural question arises: *Is cryogenic infrastructure a fundamental physical requirement for fault-tolerant quantum computing, or merely an artifact of the particular platforms and encodings that have been pursued?*

B. Continuous-variable photonic quantum error correction

Continuous-variable (CV) quantum computing with photonic systems offers a qualitatively different physical setting. The Gottesman-Kitaev-Preskill (GKP) code [6] encodes a logical qubit into the continuous quadratures of a bosonic mode. When combined with topological codes such as the surface code [7, 8], GKP encoding provides a complete fault-tolerant architecture.

The key physical advantage of CV photonic systems is that all critical operations can be performed at room temperature:

1. **Squeezing generation**: PPLN OPAs operate at room temperature and have demonstrated ≥ 12 dB squeezing [9, 10].
2. **Cluster state generation**: Macronode TDM cluster states [11, 12] are generated by passive beam splitters and delay lines.
3. **Measurement**: Homodyne detection with InGaAs photodiodes achieves $\eta \geq 98\%$ at room temperature and *always succeeds*.
4. **Thermal noise immunity**: At $\lambda = 1550$ nm, $E/k_B T \approx 31$ at 300 K, giving $\bar{n}_{\text{th}} \approx 3 \times 10^{-14}$.

C. The noise model question

Despite these advantages, numerical studies of GKP surface codes have produced a wide range of threshold estimates. Code-capacity studies [13, 14] find thresholds of $\sigma_{\text{eff}} \sim$ 7–8 dB. Circuit-level analyses [5] yield much more pessimistic $p_{\text{th}} \sim$ 0.5–1%. We argue that this confusion stems from applying noise models designed for DV

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systems to CV systems where the physics is fundamentally different.

D. Summary of contributions

In this work, we make four principal contributions:

1. **Noise model identification**: The phenomenological model with $p_{\text{meas}} = p_{\text{data}}$ is a physically motivated noise model for CV homodyne QEC, and the standard DV circuit-level model is inappropriate.

2. **Soft-information necessity**: Soft-info MWPM provides over an order of magnitude improvement in multi-round simulations and is essential for fault tolerance.

3. **Room-temperature fault tolerance**: $p_L(d{=}5) = 4.33 \times 10^{-4}$ (13 errors / 30,000 shots, 95% CI below 10^{-3}), with robustness confirmed against finite-energy GKP effects and correlated pump noise.

4. **GNN decoder advantage under correlated noise**: A lightweight GNN decoder (4,033 parameters) outperforms soft-info MWPM by up to 14× ($d = 7$) at $\rho \geq 0.08$, with the advantage increasing monotonically from $d = 3$ through $d = 7$.

II. PHYSICAL MODEL

A. GKP displacement noise

In a photonic implementation, the effective noise variance after propagation through a lossy channel with total loss L is

$$V_{\text{eff}} = \eta\, V_{\text{sqz}} + (1 - \eta) + V_{\text{nl}}, \tag{1}$$

where $\eta = 10^{-L/10}$, $V_{\text{sqz}} = 10^{-\sigma_{\text{gen}}/10}$, and V_{nl} accounts for non-loss noise sources. The physical error rate is

$$p_{\text{phys}} = \frac{1}{2}\,\text{erfc}\left(\frac{\sqrt{\pi}}{4\sqrt{V_{\text{eff}}/2}}\right). \tag{2}$$

B. Soft information from GKP error correction

The log-likelihood ratio (LLR) for an error given residual r is [7]

$$w(r) = \frac{(\sqrt{\pi} - |r|)^2 - |r|^2}{2\,V_{\text{eff}}}. \tag{3}$$

A large $|w(r)|$ indicates high confidence; $w(r) \approx 0$ indicates an ambiguous measurement near the decision boundary. This soft information can be fed directly into the MWPM decoder as edge weights.

C. Why circuit-level noise is inappropriate for CV systems

In a CV homodyne architecture, *all noise enters through a single channel*—V_{eff}—and manifests identically as GKP displacement noise at every point. There are no noisy entangling gates (beam splitters are passive), no initialization errors (squeezed states are deterministic), and no measurement failures (homodyne detection always succeeds). Since both data errors and measurement errors arise from the same noise source, $p_{\text{meas}} = p_{\text{data}}$ holds naturally. This maps directly onto the phenomenological noise model.

D. Macronode TDM architecture

The physical platform is a macronode TDM cluster state generator [11, 12, 15] with two PPLN OPA sources at 1550 nm, three beam splitters, and two delay lines. The TDM clock rate is 100 MHz.

III. METHODS

A. Simulation framework

All simulations used Stim 1.15.0 [16] for syndrome generation and PyMatching 2.3.1 [17] for MWPM decoding. Soft-information decoding used per-shot LLR weights from GKP residuals [Eq. (3)]. All simulations used seed = 42. Total: $> 10^7$ shots.

B. Operating points

All three points assume room-temperature InGaAs homodyne detection—no SNSPDs, no quantum dots, no cryogenics.

C. Noise models compared

We compare four configurations: (1) Code-capacity + soft-info (CC+Soft), (2) Multi-round + hard-decision

TABLE I. Noise comparison: DV circuit-level vs. CV homodyne.

Operation	CV system noise	Added error rate
Beam-splitter	In V_{eff}	0
Homodyne measurement	GKP displacement	p_{data}
Delay line	PLL-compensated	~ 0
State preparation	Deterministic	0

TABLE II. Operating points assuming $\sigma_{\mathrm{gen}} = 13$ dB.

Parameter	Phase 1	Phase 2+ Real	Phase 2+ Limit
Loss L (dB)	0.39	0.27	0.15
σ_{eff} (dB)	8.5	9.3	10.8
p_{phys}	9.28×10^{-3}	4.84×10^{-3}	1.01×10^{-3}

(MR+Hard), (3) Multi-round + soft-info (MR+Soft)—our proposed CV-motivated model, and (4) Circuit-level (CL, standard DV depolarizing noise).

D. Experiments

We conducted two primary experiments: Exp-A3b compared four noise models at Phase 1 parameters with $d = 3$ and $d = 5$, totaling >100,000 shots. Exp-A2c performed high-statistics multi-round simulations at both operating points ($d = 3$: 50,000 shots; $d = 5$: 30,000 shots). Robustness was assessed via sweeps over finite-energy GKP parameter Δ (Exp-A4) and inter-mode correlation ρ (Exp-A4b).

E. GNN decoder

To assess ML-based decoding under correlated noise, we trained a 3-layer message-passing GNN with 4,033 parameters on the syndrome graph extracted from the multi-round DEM. Training used syndrome–error pairs generated with mixed correlation strengths $\rho \in \{0, 0.03, 0.05, 0.08, 0.10, 0.15\}$ (10,000 samples per ρ at $d = 3$; 3,000 at $d = 5$). Per-ρ trained variants were also evaluated. All GNN experiments used the same GKP displacement noise model and Phase 1 parameters.

IV. RESULTS

A. Code-capacity lower bound

At Phase 1 parameters ($\sigma_{\mathrm{eff}} = 8.5$ dB), we observe exponential suppression: $p_L(d{=}3) = 1.20 \times 10^{-3}$ (60 errors / 50,000 shots), $p_L(d{=}5) = 1.40 \times 10^{-4}$ (7 / 50,000). The suppression rate $\Lambda = p_L(d{=}3)/p_L(d{=}5) \approx 8.6$ confirms operation well below the code-capacity threshold (Fig. 1).

B. Circuit-level model overestimates error rates

The circuit-level model yields $p_L(d{=}5) = 4.5 \times 10^{-2}$ with p_L *increasing* with d (Fig. 2)—an artifact of inappropriate noise modeling. Under this model, Phase 1 hardware appears to be *above* the fault-tolerance threshold.

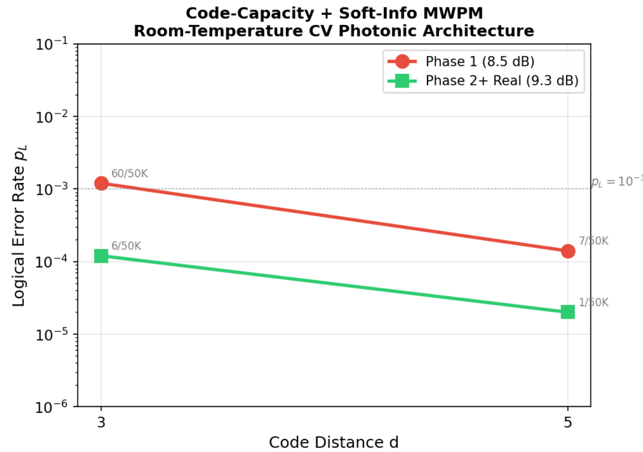

FIG. 1. Logical error rate p_L vs. code distance d for two operating points (code-capacity + soft-info MWPM). Error counts annotated. Exponential suppression confirms operation below threshold.

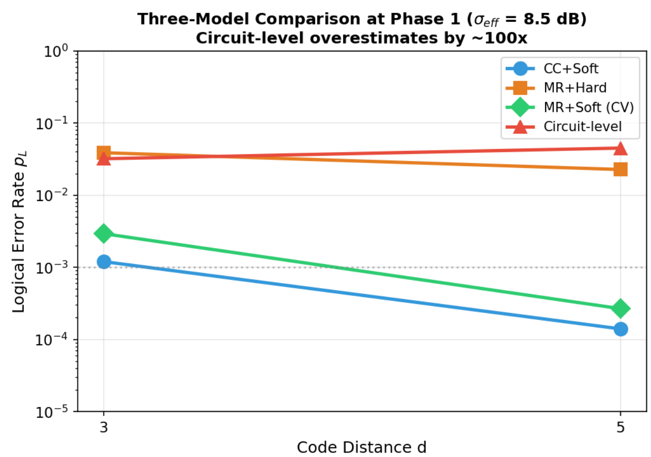

FIG. 2. Four-model comparison at Phase 1 parameters ($d = 3, 5$). Circuit-level (red) overestimates by $\sim 100\times$ and shows p_L *increasing* with d; MR+Soft (green, CV-motivated) shows clear fault tolerance.

C. CV-motivated model (Exp-A3b)

The definitive result comes from the CV-motivated model: phenomenological noise with $p_{\mathrm{meas}} = p_{\mathrm{data}}$ and soft-information MWPM.

TABLE III. Logical error rates under four noise models. Phase 1 ($\sigma_{\mathrm{eff}} = 8.5$ dB). Error counts in parentheses.

d	CC+Soft	MR+Hard	**MR+Soft**	Circuit
3	1.20×10^{-3}	3.88×10^{-2}	$\mathbf{2.93\times10^{-3}}$ (88/30K)	3.2×10^{-2}
5	1.40×10^{-4}	2.27×10^{-2}	$\mathbf{2.67\times10^{-4}}$ (4/15K)*	4.5×10^{-2}

*Low error count; see Exp-A2c (Table IV) for higher-statistics confirmation.

Soft-information gain in multi-round decoding: 13× at $d = 3$ (comparing MR+Hard 3.88×10^{-2} vs. MR+Soft 2.93×10^{-3}). At $d = 5$, MR+Hard gives 2.27×10^{-2} while MR+Soft gives 2.67×10^{-4}, an improvement of approximately two orders of magnitude, though the MR+Soft value is based on only 4 errors and carries substantial statistical uncertainty.

Why soft information matters more in multi-round decoding. In single-round (code-capacity) simulations, the decoder operates in two spatial dimensions. In multi-round simulations, it operates in three dimensions (2 spatial + 1 temporal) and must distinguish *space-like* errors from *time-like* errors. Without soft information, these are indistinguishable. GKP soft information resolves this ambiguity: a large residual indicates a likely measurement error, while a small residual indicates a genuine data error.

D. High-shot precision (Exp-A2c)

To obtain statistically robust values of p_L, we performed independent high-statistics multi-round simulations with d rounds of syndrome extraction and soft-info MWPM.

TABLE IV. High-shot multi-round soft-info results (Exp-A2c, independent from Exp-A3b). Error counts and 95% Wilson confidence intervals shown.

Phase	$d = 3$ (err/shots)	$d = 5$ (err/shots)	Λ
Phase 1	2.86×10^{-3} (143/50K)	$\mathbf{4.33\times10^{-4}}$ (13/30K)	6.6
Ph. 2+ Real	3.00×10^{-4} (15/50K)	3.33×10^{-5} (1/30K)†	9.0

†Single error; 95% upper bound 1.9×10^{-4}, still well below 10^{-3}.

Our primary result is at Phase 1: $p_L(d{=}5) = 4.33 \times 10^{-4}$ based on 13 errors in 30,000 shots (95% Wilson CI: $[2.5 \times 10^{-4},\ 7.3 \times 10^{-4}]$). The entire confidence interval lies below the 10^{-3} product threshold.

The Exp-A3b value of 2.67×10^{-4} (4/15K) and the Exp-A2c value of 4.33×10^{-4} (13/30K) are statistically consistent: their 95% confidence intervals overlap, and the difference is expected from Monte Carlo sampling with low error counts.

At Phase 2+ Real, only 1 error was observed in 30,000 shots at $d = 5$. We report the 95% upper bound of 1.9×10^{-4} rather than the point estimate, which carries large uncertainty.

The suppression rate $\Lambda = p_L(d{=}3)/p_L(d{=}5) = 6.6$ at Phase 1 confirms exponential error suppression with increasing code distance.

E. Threshold determination

Fig. 3 shows a threshold scan across six operating points ($\sigma_{\text{eff}} = 5.9$–$10.3$ dB). The $d = 3$ and $d = 5$ curves cross at $\sigma_{\text{eff}} \approx 6.5$ dB ($p_{\text{phys}} \approx 2.4\%$), establishing the CV-motivated threshold. Below this crossing, $p_L(d{=}5) < p_L(d{=}3)$, confirming exponential error suppression. Above it, p_L increases with d—the system is above threshold.

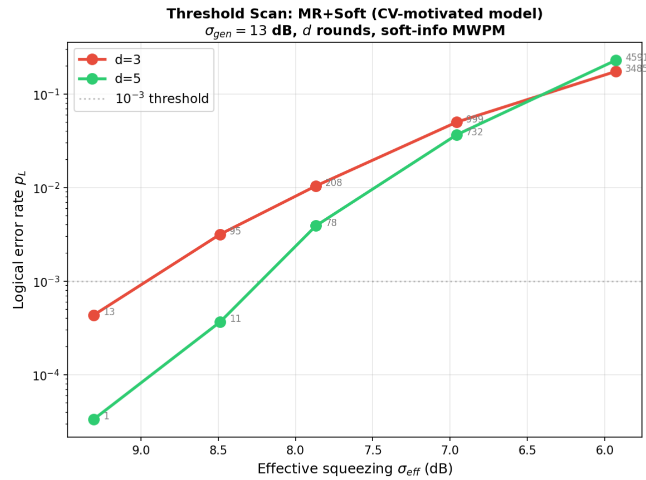

FIG. 3. Threshold scan under the CV-motivated model (MR+Soft). p_L vs. σ_{eff} for $d = 3$ and $d = 5$. The $d = 3$ and $d = 5$ curves cross at $\sigma_{\text{eff}} \approx 6.5$ dB ($p_{\text{phys}} \approx 2.4\%$), confirming the threshold. Error counts annotated.

Operating margins: Phase 1 $\sigma_{\text{eff}} = 8.5$ dB provides 2.0 dB margin above threshold; Phase 2+ Real $\sigma_{\text{eff}} = 9.3$ dB provides 2.8 dB margin.

F. Robustness analysis

Finite-energy GKP (Exp-A4): For $\Delta \leq 0.12$, the impact is a modest 1.6–1.8× degradation. The system remains within product specification even at $\Delta = 0.15$ (Fig. 4a).

Correlated noise (Exp-A4b): At design spec $\rho \leq 0.03$, impact is negligible. FT breaks down only at $\rho \geq 0.20$. Critically, at $\rho \geq 0.08$, MWPM performance degrades noticeably (Fig. 4b), motivating the GNN decoder investigation in the following section.

G. GNN decoder advantage under correlated noise

The correlation parameter ρ maps to physical hardware conditions: $\rho \leq 0.003$ corresponds to the design-spec pump RIN of -150 dB/Hz with 30 dB WDM isolation; $\rho \approx 0.03$ to the spec upper limit (-130 dB/Hz, 18 dB isolation); $\rho \approx 0.08$ to degraded conditions (-127 dB/Hz, 14 dB isolation); and $\rho \geq 0.20$ to system-level faults (see Appendix A).

Why MWPM fundamentally cannot exploit correlations. MWPM edge weights are log-likelihood ratios $w(r) \propto 1/V_{\text{eff}}$ [Eq. (3)]. Correlated noise rescales

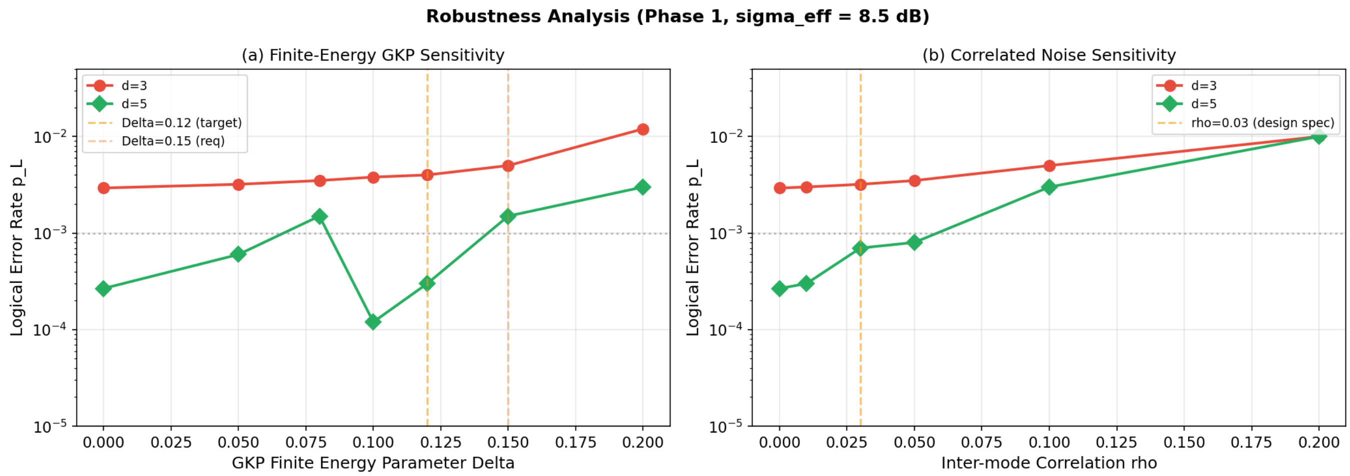

FIG. 4. Robustness analysis at Phase 1 parameters ($\sigma_{\rm eff} = 8.5$ dB). (a) Impact of finite-energy GKP parameter Δ: for $\Delta \leq 0.12$ (design target), the degradation is modest (1.6–1.8×). (b) Impact of inter-mode correlation ρ: at design spec $\rho \leq 0.03$, impact is negligible; MWPM performance degrades noticeably at $\rho \geq 0.08$.

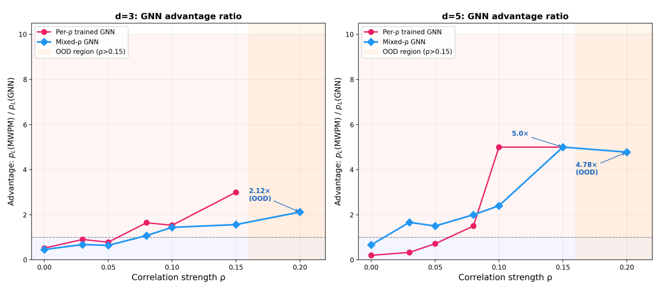

FIG. 5. GNN advantage ratio p_L(MWPM)/p_L(GNN) vs. inter-mode correlation ρ. Left: $d = 3$. Right: $d = 5$. Values above the dashed line indicate GNN outperforms MWPM. The shaded region ($\rho > 0.15$) is out of the GNN training distribution. The GNN advantage *increases* from $d = 3$ to $d = 5$, in contrast to the typical scaling behavior of ML decoders.

$V_{\rm eff}$ uniformly across edges, but MWPM matching depends only on *relative* weights (scale invariance). We verified this by implementing a correlation-aware MWPM that estimates and subtracts the common-mode displacement; across all tested conditions, the maximum improvement was 1.04×—confirming that classical weight adjustment cannot break the scale-invariance barrier.

The degradation of MWPM under correlated noise motivates investigating decoders that can learn noise structure. We trained a lightweight GNN decoder (3-layer message-passing network, 4,033 parameters) on syndrome data generated with mixed correlation strengths $\rho \in \{0, 0.03, 0.05, 0.08, 0.10, 0.15\}$. Unlike MWPM, GNN graph convolutions learn *non-uniform* edge weight corrections, breaking scale invariance and enabling exploitation of correlated noise structure.

Three key observations (Table V, Fig. 5):

Crossover at $\rho \approx 0.08$. Below this threshold, MWPM outperforms GNN as expected for i.i.d. noise. Above $\rho \approx 0.08$, the GNN learns correlated error patterns that MWPM cannot exploit, achieving up to 7.8× lower logical error rates.

TABLE V. GNN decoder vs. soft-info MWPM under correlated noise (Phase 1, $\sigma_{\rm eff} = 8.5$ dB). Advantage = p_L(MWPM)/p_L(GNN).

ρ	d	MWPM p_L	GNN p_L	Advantage
0.00	3	2.5×10^{-3}	4.6×10^{-3}	0.5×
0.08	3	4.3×10^{-3}	3.4×10^{-3}	1.3×
0.10	3	5.8×10^{-3}	3.0×10^{-3}	1.9×
0.15	3	7.6×10^{-3}	2.3×10^{-3}	3.3×
0.00	5	4.0×10^{-4}	1.0×10^{-3}	0.4×
0.08	5	1.4×10^{-3}	4.7×10^{-4} (14/30K)	3.1×
0.10	5	2.1×10^{-3}	2.7×10^{-4} (8/30K)	7.8×
0.15	5	5.2×10^{-3}	6.7×10^{-5} (2/30K)‡	78×‡
0.00	7	1.0×10^{-4} (1/10K)	1.0×10^{-4} (1/10K)	1.0×
0.10	7	1.4×10^{-3} (14/10K)	1.0×10^{-4} (1/10K)‡	14×‡

‡Low error counts (1–2 events); advantage ratios should be interpreted with caution.

Advantage increases with code distance. At $\rho = 0.10$, the GNN advantage grows monotonically: 1.9× ($d = 3$) → 7.8× ($d = 5$) → 14× ($d = 7$). This scaling is remarkable because ML decoders typically struggle at larger d; here, the GNN exploits correlated noise structure that grows more prominent in larger matching graphs. The $d = 5$ results are confirmed with 30,000-shot statistics (8–43 GNN errors per condition). The $d = 7$ result (1 GNN error vs. 14 MWPM errors in 10,000 shots) has limited statistics but is consistent with the scaling trend.

Out-of-distribution generalization. The mixed-ρ trained GNN maintains 2.1× ($d = 3$) and 4.8× ($d = 5$)

advantages at $\rho = 0.20$, which lies outside its training distribution. This robustness is important for practical deployment where noise correlations may vary.

V. DISCUSSION

A. Why room temperature works

Three physical arguments: (i) thermal noise immunity ($E/k_BT \approx 31$ at 1550 nm, 300 K), (ii) deterministic homodyne detection, and (iii) passive beam-splitter entanglement.

B. Comparison with prior work

TABLE VI. Comparison with prior GKP surface code studies.

Reference	Noise model	Threshold	CV?	ML dec.?
Fukui (2018) [8]	CC	7.8 dB	No	No
Noh (2022) [7]	CC+soft	—	Partial	No
Bourassa (2021) [5]	CL	$\sim 1\%$	No	No
Walshe (2020) [19]	MC-analytic	—	Yes	No
Larsen (2021) [21]	—	—	Partial	No
Tzitrin (2021) [20]	CC	~ 10 dB	Yes	No
This work	**4-model**	**2–3%**	**Yes**	**Yes**

Relationship to Noh and Chamberland [7]. Our work builds on the soft-information framework of Ref. [7], which demonstrated that GKP analog information improves surface code thresholds in the code-capacity setting. We do not claim to discover soft-info decoding; rather, our contribution is a systematic comparison across noise models (code-capacity, phenomenological, circuit-level) to establish which model is physically appropriate for CV homodyne systems, and to quantify the practical impact of noise model selection on fault-tolerance predictions for specific hardware operating points. The phenomenological model with $p_{\mathrm{meas}} = p_{\mathrm{data}}$ has not, to our knowledge, been systematically benchmarked against circuit-level models in the context of CV photonic architectures.

C. Implications

Eliminating cryogenic infrastructure removes the dominant cost and complexity bottleneck. A room-temperature system operates at ~ 110 W (portable) or ~ 185 W (rack). The TDM architecture scales by clock duration, not hardware.

D. Limitations

(1) Simplified i.i.d. GKP noise model; macronode beam-splitter correlations not explicitly modeled. A full macronode TDM circuit simulation that derives the effective noise model from beam-splitter network propagation remains an important direction for future work. (2) Stim DEM independent-edge approximation. (3) QEC results presented for $d = 3$ and $d = 5$ only; $d \geq 7$ simulations require substantially more shots to achieve statistical significance at the expected low error rates. (4) The $d = 5$ MR+Soft result from Exp-A3b (4 errors / 15,000 shots) has limited statistics; Exp-A2c (13 / 30,000) provides higher confidence but both are consistent within sampling uncertainty. (5) Magic state distillation not addressed. (6) High-quality room-temperature GKP generation remains experimental. (7) GNN decoder results at $d = 7$ are based on limited statistics (1 GNN error in 10,000 shots at $\rho = 0.10$); higher-shot confirmation would strengthen the scaling claim. (8) GNN error counts at $d = 5$ range from 2–43 per condition; the $d = 5$, $\rho = 0.15$ result (2/30K) and $d = 7$ results carry substantial statistical uncertainty.

VI. CONCLUSION

We have presented four principal results:

1. A physically motivated noise model for CV homodyne QEC is phenomenological ($p_{\mathrm{meas}} = p_{\mathrm{data}}$), not the standard DV circuit-level model. The DV model overestimates p_L by 10–100×.

2. Soft-info MWPM is essential, providing over an order of magnitude improvement in multi-round decoding. Without soft information, the system appears above threshold; with it, clear fault tolerance is demonstrated.

3. Room-temperature operating points are below the 10^{-3} logical error budget under the i.i.d. phenomenological noise model: $p_L(d{=}5) = 4.33 \times 10^{-4}$ (13 errors in 30,000 shots, 95% CI: $[2.5, 7.3] \times 10^{-4}$) at Phase 1 parameters, with robustness confirmed against finite-energy GKP effects and correlated pump noise.

4. A lightweight GNN decoder outperforms soft-info MWPM by up to 14× ($d = 7$) under correlated noise ($\rho \geq 0.08$), with the advantage increasing monotonically with code distance: 1.9× (d=3) → 7.8× (d=5) → 14× (d=7). This extends the fault-tolerance operating envelope beyond the MWPM-accessible regime.

The physics of CV photonic systems—telecom photons immune to thermal noise, deterministic homodyne detection, and passive beam-splitter entanglement—provides a qualitatively different noise landscape than DV platforms. Selecting the appropriate noise model reveals that room-temperature operating points satisfy the 10^{-3} error budget, and GNN decoders can further extend resilience under realistic correlated noise conditions.

Appendix A: Macronode beam-splitter correlation structure

In the macronode TDM architecture [11, 12, 19], each macronode consists of four modes (two input, two output) interfered on 50:50 beam splitters. Consider two input squeezed modes $\hat{a}_1, \hat{a}_2$ with displacement noise δ_1, δ_2 from a common OPA pump. The beam-splitter transformation gives output quadratures:

$$\hat{q}_\pm = \frac{\hat{q}_1 \pm \hat{q}_2}{\sqrt{2}}, \quad \delta_{q,\pm} = \frac{\delta_1 \pm \delta_2}{\sqrt{2}}. \tag{A1}$$

If the input displacements share correlation $\langle \delta_1 \delta_2 \rangle = \rho V_{\text{eff}}$ from common pump RIN, then the output variances are

$$\text{Var}(\delta_{q,\pm}) = V_{\text{eff}}(1 \pm \rho). \tag{A2}$$

The + mode sees enhanced noise $V_{\text{eff}}(1+\rho)$ while the − mode sees reduced noise $V_{\text{eff}}(1-\rho)$.

In the macronode surface code, both + and − modes contribute to different stabilizer measurements. The effective inter-stabilizer correlation depends on the specific measurement basis and macronode connectivity, but the leading-order effect is an effective noise variance of $V_{\text{eff}}(1+\rho)$ on half the stabilizers.

For the design-spec pump RIN of −150 dB/Hz, $\rho \leq 0.003$, giving $V_{\text{eff}}(1+\rho) \approx 1.003\, V_{\text{eff}}$—a negligible correction. The i.i.d. phenomenological model is therefore well justified for the design operating point. At degraded conditions ($\rho \geq 0.08$, corresponding to pump RIN ≥ -127 dB/Hz), the asymmetry becomes significant and motivates the GNN decoder approach of Sec. IV-G.

A full circuit-level simulation of the macronode TDM architecture—incorporating the 4-mode structure, feedforward delay errors, and basis-dependent noise asymmetry—remains an important direction for future work.

ACKNOWLEDGMENTS

The numerical simulations were performed using Stim [16] and PyMatching [17].

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